

EXTENDED READING AND STATIC VERIFICATION

Better advice, weaker learning?

AI, Human Cognition and Knowledge Collapse

Acemoglu, Kong and Ozdaglar

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github.com/gsaco/ai-04-acemoglu

Static derivations and an original assumption test.

Dynamic and welfare results are reported from the paper, not reproduced.

The bridge: private decisions create public information

The information that saves effort today can weaken learning tomorrow.

- A person needs general understanding and context-specific information.
- Human effort improves their own information and leaves a public trace.
- Personalized AI replaces part of the private reason to learn.
- That response can reduce the public knowledge inherited by later cohorts.

Introduction; §2, pp. 5–6: the contribution combines individual information acquisition with collective learning, rather than treating collaboration as a single isolated task.

Production: complementarity is an assumption

Let $b_G, b_I \in \{0, 1\}$ indicate prediction errors within a unit band.

$$f(b_G, b_I) = f(0, 0) + \Delta_G b_G + \Delta_I b_I + \Delta_X b_G b_I.$$

$$\Delta_G = f(1, 0) - f(0, 0),$$

$$\Delta_I = f(0, 1) - f(0, 0),$$

$$\Delta_X = f(1, 1) - f(1, 0) - f(0, 1) + f(0, 0).$$

Monotone f ; $\Delta_G + \Delta_I + \Delta_X = 1$.

Assumption 1: $\Delta_I = 0$, $\Delta_X > 0$. General knowledge is essential for context-specific information to pay off.

§3.1, p. 8. Leontief is one example ($\Delta_G = \Delta_I = 0$, $\Delta_X = 1$), not the entire model.

Normal information: precisions add

The common state drifts; the individual state is fresh each period:

$$\theta_{t+1} = \theta_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \Sigma^2), \quad \theta_{i,t} \sim N(0, \sigma^2).$$

Independent human and AI signals about the individual state give

$$Y = \sigma^{-2} + \lambda_I e + \tau_A.$$

Public information: $X = \text{Var}(\theta_t \mid \mathcal{I}_t)^{-1}$. It is inherited when effort is chosen.

Posterior means maximize the probability of lying within the symmetric unit-tolerance band. The two success probabilities are $G(X)$ and $G(Y)$.

§§3.1–3.3, pp. 7–12, eqs. (4)–(5). Gaussian, conditionally independent information.

The static objective and its unique optimum

Dropping $f(0,0)$, and using $\Delta_I = 0$:

$$U = G(X)\Delta_G + G(X)G(Y)\Delta_X - e^\alpha/\alpha.$$

$$\Delta_X G(X)\lambda_I g'(Y) = e^{\alpha-1}.$$

For $e > 0$,

$$U_{ee} = \Delta_X G(X)\lambda_I^2 g'(Y) - (\alpha - 1)e^{\alpha-2} < 0.$$

At $X > 0$, marginal benefit at zero is positive, while marginal cost ultimately dominates: a unique, finite, interior optimum exists.

§3.5, p. 14. Conditions: $\alpha > 1$, $\lambda_I > 0$, $\Delta_X > 0$,

$\sigma^{-2} > 0$, finite $\tau_A \geq 0$. At $X = 0$, the optimum is the boundary $e = 0$.

Deriving the two signs by hand

$$G(z) = 2\Phi(\sqrt{z}) - 1, \quad g(z) = \frac{e^{-z/2}}{\sqrt{2\pi z}}.$$

Log-differentiation makes the curvature transparent:

$$\frac{g'(z)}{g(z)} = -\frac{1}{2} - \frac{1}{2z} < 0.$$

$$U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0,$$

$$U_{e\tau_A} = \Delta_X G(X) \lambda_I g'(Y) < 0.$$

Interpretation: X increases the value of private precision; τ_A reduces its marginal value by already supplying it.

Observation 1, p. 13. Strict signs on $X, Y > 0$ with $\Delta_X, \lambda_I > 0$;

short-lived atomistic agents do not internalize their public learning contribution.

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From payoff cross-partial to effort responses

Differentiate the interior FOC $U_e(e^*, X, \tau_A) = 0$:

$$e_X^* = -\frac{U_{eX}}{U_{ee}} > 0, \quad e_{\tau_A}^* = -\frac{U_{e\tau_A}}{U_{ee}} < 0.$$

The denominator matters: substitution is a choice result because the optimum is concave.

- These derivatives hold at fixed inherited X or fixed τ_A .
- Aggregation capacity I is absent from the current private FOC.
- I matters through the public information later cohorts inherit.

Observation 2, p. 14; own implicit-function derivation.

Do not confuse the technology parameter λ_I with island size I .

The Bayesian bridge, read without solving dynamics

The paper's public-information recursion is

$$V_{t+1} = \frac{1}{X_t + \lambda_G l e_t} + \Sigma^2, \quad X_{t+1} = \left[\frac{1}{X_t + \lambda_G l e_t} + \Sigma^2 \right]^{-1}.$$

- Pooling human signals adds **precision**: $\lambda_G l e_t$.
- State innovation adds **variance**: Σ^2 .
- The static best response supplies $e_t = e(X_t, \tau_A)$.

§3.2, eq. (3), p. 11; Proposition 1, p. 15.

Reported model equation only: no recursion simulation or steady-state reconstruction.

What the dynamic results claim

Assumption 1; $I, \lambda_I, \lambda_G, \Sigma^2, \sigma^2 > 0$; finite $\tau_A \geq 0$.

Cost regime	Reported result
$\alpha - 1 > 1/4$	Zero is unstable; one positive stable steady state attracts every $X_1 > 0$. Zero remains a fixed point.
$0 < \alpha - 1 < 1/4$	Below τ_A^c , zero and a high state are stable, separated by $\bar{X}_m$. Above τ_A^c , only zero remains.

In the second regime, $X_1 > \bar{X}_m$ selects the high state; $X_1 < \bar{X}_m$ selects collapse. At $X_1 = \bar{X}_m$, the state stays there.

Boundary discipline: these claims exclude $\alpha - 1 = 1/4$. We use strict inequalities around τ_A^c ; the paper's treatment of equality is inconsistent.

Propositions 3 and 5, pp. 18, 20; see extra/analysis.md for the threshold caveat.

Welfare: which quantity is being differentiated?

Static optimized utility $v(X, \tau_A) = \max_e U(e; X, \tau_A)$:

$$\left. \frac{\partial v}{\partial \tau_A} \right|_X = G(X) \Delta_X g(Y) > 0.$$

High-steady-state welfare, as reported in the paper:

$$\frac{d\bar{U}^+}{d\tau_A} = \underbrace{G(\bar{X}_h) \Delta_X g(\bar{Y}_h)}_{\text{direct gain}} + \underbrace{g(\bar{X}_h) [\Delta_G + G(\bar{Y}_h) \Delta_X]}_{\text{loss of public knowledge}} \frac{d\bar{X}_h}{d\tau_A}.$$

Private improvement at fixed knowledge does not imply a social long-run improvement.

Static envelope calculation: own verification. Long-run decomposition: §4.3, p. 25.

Welfare is output minus effort cost, relative to $f(0,0)$; dynamic results are read-only.

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Welfare theorems: the qualifications are the result

Common conditions: baseline model and Assumption 2, $\sigma^{-2} \geq \sqrt{2} - 1$; other parameters held fixed.

Proposition 10: if $\alpha - 1 > 1/4$, a finite $\tau_A^* \geq 0$ maximizes $\bar{U}^+$. Welfare increases below it, decreases above it, and tends to zero as $\tau_A \rightarrow \infty$. Realized high-state welfare requires $X_1 > 0$.

Proposition 11: if $0 < \alpha - 1 < 1/4$ and the positive branch exists, $0 \leq \tau_A^* < \tau_A^c$. Welfare on that branch increases then decreases; above τ_A^c the collapse outcome has zero normalized welfare. Realized welfare also depends on the initial basin.

Not necessarily interior: $\tau_A^* = 0$ is explicitly allowed. Without Assumption 2, the paper discusses failure of single-peakedness.

§4.1, pp. 23–24; Assumption 2 and footnote 6, p. 25; Propositions 10–11, p. 26.

When no positive branch exists even at zero AI precision, do not assert a positive-branch optimum.

Section 5 changes information production

Extension	What changes and what is claimed
5.1 Aggregation	$I(\tau_A) = I_0 + e^{\eta\tau_A}$. For $\eta < 1/[2(\alpha - 1)]$, knowledge still tends to zero at very high AI precision.
5.2 Synthetic data	Adds $0 < \tau_{\text{syn}} < \infty$ public precision each period. Least and greatest steady states are positive; effort and knowledge fall with τ_A .
5.3 Effort bundling	Public precision scales with e^β . For $\beta > 0$, the stability comparison becomes $\alpha - 1 \gtrless \beta/4$. $\beta = 0$ is exceptional.

Never relaxed: $\Delta_I = 0$. Section 5.3 changes the public-learning technology; it does not give context-specific knowledge standalone output value.

§5, Propositions 14–16, pp. 29–32. The formal §5.3 extension has one effort choice, not a separately optimized allocation between two effort types.

My extension: give context-specific knowledge some value

Allow $\Delta_I > 0$, retain $\Delta_X > 0$, and preserve normalization:

$$U = \Delta_G G(X) + [\Delta_I + \Delta_X G(X)]G(Y) - e^\alpha / \alpha.$$

$$U_{eX} = \Delta_X \lambda_I g(X)g(Y) > 0,$$

$$U_{e\tau_A} = [\Delta_I + \Delta_X G(X)]\lambda_I g'(Y) < 0.$$

At the zero-knowledge boundary and finite τ_A ,

$$U_e(0; 0, \tau_A) = \Delta_I \lambda_I g(\sigma^{-2} + \tau_A) > 0.$$

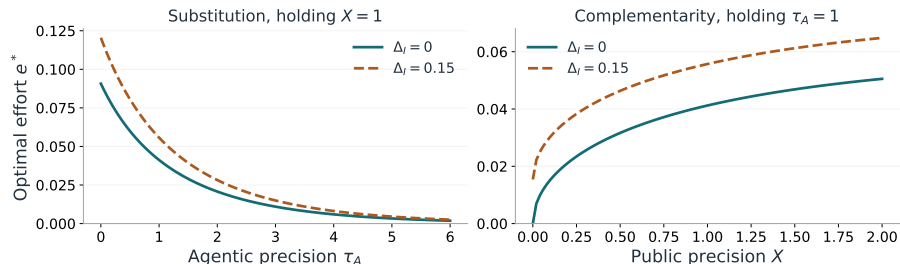
Verdict: both signs survive; zero effort at zero knowledge does not. This tests the essential-input assumption, without solving a new dynamic model.

Own static extension, not a result stated by the authors.

$\alpha > 1$, $\lambda_I > 0$, $\sigma^{-2} > 0$, $\Delta_I > 0$; choose $\Delta_G \geq 0$ to keep monotonicity.

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Reproducible evidence, not a calibration



Six symbolic identities; 444 numerical optima; FOC residuals below 2.5×10^{-15} . Implicit-function and envelope derivatives match finite differences.

At $X = 0$, $\tau_A = 1$: $e^* = 0$ under $\Delta_I = 0$ versus 0.01539 under $\Delta_I = .15$.

extra/static_checks.py; $\alpha = 2$, $\lambda_I = \sigma^{-2} = 1$, $\Delta_X = .6$, $\Delta_G = 1 - \Delta_I - \Delta_X$.

Illustrative parameters only. No dynamic paths, collapse thresholds or welfare curves fitted.

What I will verify on paper

Handwritten photo pending

Derive g' , U_{eX} and U_{eTA} on paper.

Photograph your work and save it as
`hand/derivation.jpg`.

This panel is not evidence of handwritten
work.

Write and check:

1. Differentiate G and g .
2. Derive the FOC and both signs.
3. Add $\Delta_I G(Y)$ and evaluate the boundary marginal return.

Own verdict to defend:

The static crowd-out mechanism is robust to $\Delta_I > 0$. Exact zero-knowledge collapse and zero welfare rely on additional assumptions.

Handwritten evidence remains pending until the student's real photo is supplied and the decks recompiled.

Limits, policy claims and questions for the oral

Aggregation: Proposition 9 raises high-state welfare with I , when that state exists. Better aggregation and better agentic advice are different changes.

Policy, read-only: Proposition 13 optimizes eventual Gaussian signal caps under long-run average welfare, $0 < \alpha - 1 < 1/4$, Assumption 2, and $X_1 > \bar{X}_m(0)$. It is not an unrestricted or discounted policy theorem.

Questions to defend:

- Why does zero effort follow at $X = 0$? Because $\Delta_I = 0$.
- Does $\Delta_I > 0$ reverse crowd-out? No, the negative cross-partial remains.
- Does the example prove AI always improves long-run welfare? No.
- Is an interior accuracy cap guaranteed? No; $\tau_A^* = 0$ is allowed.

Primary source throughout: supplied w34910.pdf, NBER WP 34910 (February 2026).

Detailed conditions, derivations, evidence map and five-minute script: [extra/analysis.md](#) and [extra/oral-notes.md](#).

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